Weishu Zhang

626-566-4310 | weszhang@berkeley.edu | LinkedIn | GitHub | Website

Infrastructure-leaning engineer building distributed training, serving, and production agent runtimes.

Education

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering and Computer Sciences — *May 2027*
- GPA: 4.0/4.0
- Relevant Coursework: Efficient Algorithms, Network Science, Artificial Intelligence

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java, Go, C/C++
Full Stack: React, Next.js, Node.js, FastAPI, PostgreSQL, Redis, SQLite, REST APIs
AI & Agents: LangGraph, MCP, LLM APIs, agent runtimes, structured generation, PyTorch, vLLM, QLoRA
Infrastructure: Docker, Kubernetes, gVisor, OCI, OpenTelemetry, pytest, distributed jobs, CI/CD

Experience

Amazon — Culver City, CA
Software Development Engineer Intern, AI Studio — *Sep 2026 – Nov 2026*
- Built a unified inference and training cluster supporting generative AI workloads for Amazon AI Studio.
- Designed a zero-shot generation workflow for VFX background generation and harmonization, accelerating production and lowering generation cost for VFX creators.
- Preserved spatial consistency and matched scene lighting to live-action footage for professional VFX production.

ASML — San Jose, CA
Software Engineering, Test Infrastructure | *Blog* — *May 2026 – Aug 2026*
- Built RRI to automate daily SQA regression review across Confluence, Jira, and qTest, cutting 80% of repetitive review work.
- Built a test-plan agent over 20+ coverage types, cutting plan turnaround from 2 weeks of writing to hours of review.

Ludus — Berkeley, CA
Co-Founder & Software Engineer, Cloud Agent Infrastructure — *Apr 2026 – Aug 2026*
- Co-founded an agent workspace that reached the **top 10% of Y Combinator** applicants and the **top 5% of a16z Speedrun** applicants. Multiple agents complete the same task independently from shared materials and requirements, and a reviewer can quickly switch across results to compare and preview the outcome.
- Adapted **git worktrees** so each agent session gets its own checkout, virtualenv, and process. Sessions share one object store and cannot see each other's files, packages, or uncommitted edits.
- Capped each session process in CPU and memory. An extra session adds a checkout and a virtualenv, at about **10**× lower memory overhead than a per-agent container sandbox.

Berkeley Artificial Intelligence Research (BAIR) — Berkeley, CA
Undergraduate Research Assistant — *Mar 2026 – May 2026*
- Led expert task acquisition for Agents' Last Exam, recruiting 30+ domain experts and contributing to the collection of 80+ complex, cross-domain agent tasks.
- Built an LLM-powered proposal agent that converts rough expert ideas into structured benchmark tasks, supporting iterative refinement, constraint checking, and higher-quality submissions.

University of California, San Francisco — San Francisco, CA
Software Research Assistant — *Aug 2025 – Jan 2026*
- Served fixed-schema clinical extraction for **Qwen3-32B** on one **RTX A6000** with AWQ W4A16 Marlin, a cached schema prefix, and JSON-schema decoding.
- QLoRA-tuned it on a 4,000+-example set with thinking disabled, and measured valid documents/s separately from field accuracy.

Publications

Agents' Last Exam | Co-author | arXiv:2606.05405 — NeurIPS 2026

Open Source & Projects

PyTorch | *torch.compile, torch.export, Dynamo* PRs
- Contributed production fixes to pytorch/pytorch across compiler and export internals, resolving eager-versus-compiled correctness gaps, tracing failures, and export regressions with targeted tests and maintainer-reviewed patches.

mini-vLLM | *Python, PagedAttention, KV cache, Scheduler* Code
- Built a GPU-free from-scratch model of vLLM's systems core: paged KV cache, continuous batching, preemption, and prefix caching, with reproducible benchmarks (7x more active sequences, 1.53x throughput, 54x lower p99 TTFT).